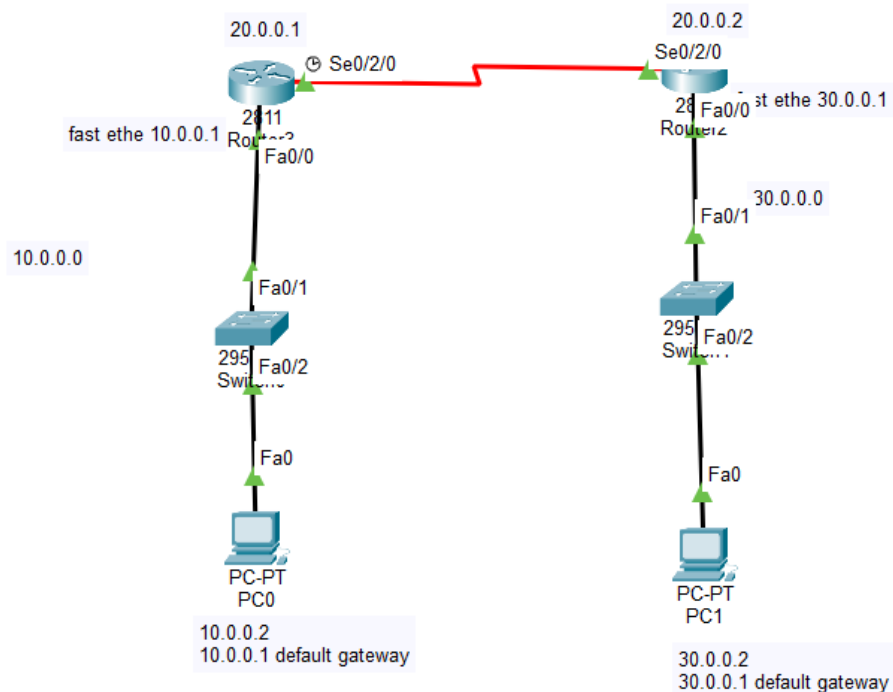


Day 1:

Static Routing Concepts Theory :What is static routing?Advantages and disadvantages of static routes.Practical :Configure static routes between two routers in Cisco Packet Tracer.

Task 1.

Open Cisco Packet Tracer and set up a simple network with two routers connected via a serial cable, assigning IP addresses to both router interfaces.



Task 2.

On Router1 in Cisco Packet Tracer, configure a static route so it can reach a PC connected to Router2's LAN network. Test connectivity using the ping command from Router1 to the PC.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 30.0.0.2

Pinging 30.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 30.0.0.2: bytes=32 time=1ms TTL=126
Reply from 30.0.0.2: bytes=32 time=5ms TTL=126
Reply from 30.0.0.2: bytes=32 time=6ms TTL=126

Ping statistics for 30.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 6ms, Average = 4ms

C:\>ping 30.0.0.2

Pinging 30.0.0.2 with 32 bytes of data:

Reply from 30.0.0.2: bytes=32 time=19ms TTL=126
Reply from 30.0.0.2: bytes=32 time=6ms TTL=126
Reply from 30.0.0.2: bytes=32 time=6ms TTL=126
Reply from 30.0.0.2: bytes=32 time=6ms TTL=126

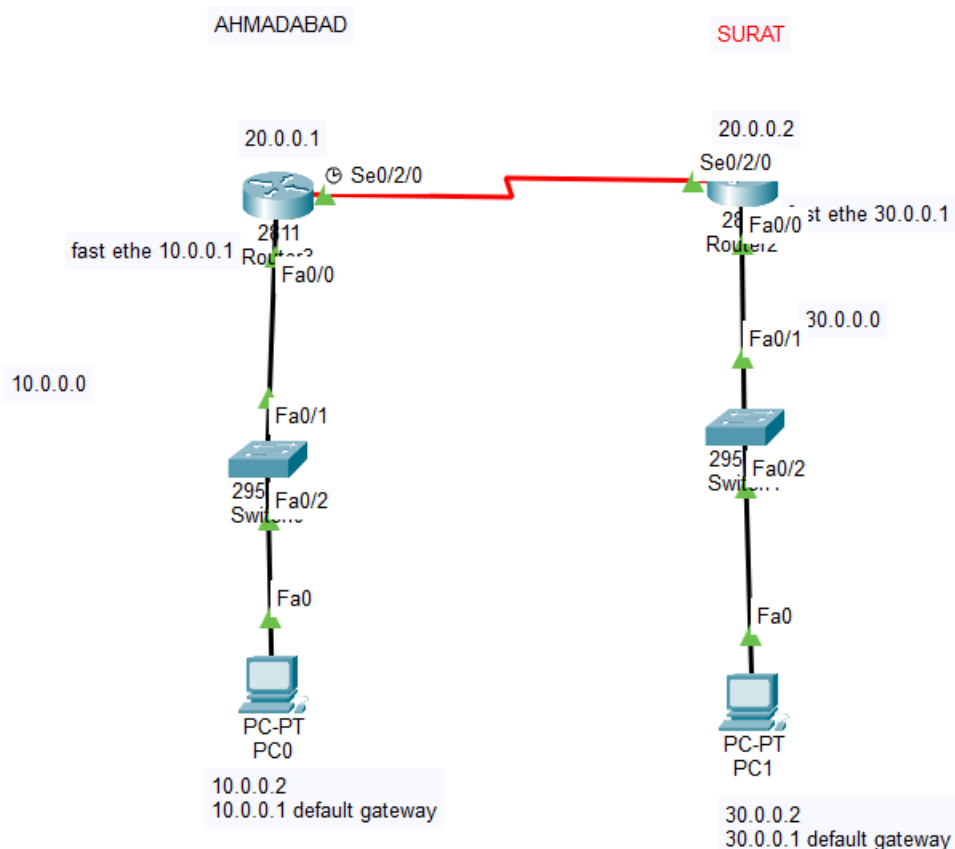
Ping statistics for 30.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 19ms, Average = 9ms

C:\>

```

Task 3.

Simulate a Zomato-style delivery network: imagine Router1 represents an Ahmedabad branch and Router2 represents a Surat branch. Configure static routes so that devices in each city's LAN can communicate with each other through the routers.



Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 30.0.0.2

Pinging 30.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 30.0.0.2: bytes=32 time=1ms TTL=126
Reply from 30.0.0.2: bytes=32 time=5ms TTL=126
Reply from 30.0.0.2: bytes=32 time=6ms TTL=126

Ping statistics for 30.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 6ms, Average = 4ms

C:\>ping 30.0.0.2

Pinging 30.0.0.2 with 32 bytes of data:

Reply from 30.0.0.2: bytes=32 time=19ms TTL=126
Reply from 30.0.0.2: bytes=32 time=6ms TTL=126
Reply from 30.0.0.2: bytes=32 time=6ms TTL=126
Reply from 30.0.0.2: bytes=32 time=6ms TTL=126

Ping statistics for 30.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 19ms, Average = 9ms

C:\>|
```

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 20.0.0.1

Pinging 20.0.0.1 with 32 bytes of data:

Reply from 20.0.0.1: bytes=32 time=9ms TTL=254
Reply from 20.0.0.1: bytes=32 time=13ms TTL=254
Reply from 20.0.0.1: bytes=32 time=1ms TTL=254
Reply from 20.0.0.1: bytes=32 time=1ms TTL=254

Ping statistics for 20.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 13ms, Average = 6ms

C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time=8ms TTL=126
Reply from 10.0.0.2: bytes=32 time=6ms TTL=126
Reply from 10.0.0.2: bytes=32 time=5ms TTL=126
Reply from 10.0.0.2: bytes=32 time=5ms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 8ms, Average = 6ms

C:\>|
```

Task 4.

List two advantages and two disadvantages of static routing compared to dynamic routing, using examples from real-life apps like IRCTC or Swiggy to illustrate your points.

Hint: Think about how static routes might impact delivery or booking updates in these apps.

Advantages	Disadvantages
1. Simple and predictable: The administrator manually decides the path. For example, an IRCTC branch network could have a fixed route to a booking server.	1. Manual configuration: If a network link fails, the administrator may need to manually change the route.
2. Low resource usage: Static routes don't exchange routing updates, so they consume very little bandwidth and CPU.	2. Poor scalability: In a large Swiggy-style network with many cities and delivery locations, configuring routes manually for every network becomes difficult.